
HACKTOBERFEST HACK DAY BENGALURU '26

OFFICIAL PROBLEM STATEMENTS & PARTICIPANT GUIDELINES

IEEE Computational Intelligence Society (IEEE CIS)
Ramaiah Institute of Technology, Bengaluru

TRACK 1

Best Use of Gemma 4

PS 01 — Multimodal Community Intelligence

Build a multimodal AI system that understands real-world information provided through images, documents, and text, and converts it into useful, actionable information for a community. Go beyond simple question-answering by combining multimodal AI with relevant real-world data or cloud-based resources.

Possible Directions

- Understand notices, forms, posters, or public documents.
- Extract important information from images and combine visual and textual context.
- Convert complex information into clear, actionable steps.
- Address a local or community use case.
- Enrich the solution using public datasets, APIs, or AWS resources where useful.

Core Challenge: Make Gemma 4's multimodal capabilities central to a practical workflow. AWS resources are encouraged but optional.

PS 02 — Intelligent Personal / Professional Copilot

Build an AI copilot that understands a user's unstructured inputs, reasons about them, and takes or recommends a sequence of actions. The system must demonstrate more than a single AI response.

Possible Capabilities

- Multi-step reasoning and task planning.
- Document or image understanding, tool usage, and structured outputs.
- Context or memory and integration with external services.
- Human approval before important or consequential actions.

Core Challenge: Design a reliable AI workflow in which Gemma 4 plays a meaningful role in understanding and decision-making. The use case, architecture, and technology stack are open-ended.

TRACK 2

Best Open-Source AI Project

PS 03 — Open-Source AI Agent for Real-World Operations

Build an AI agent that takes a real-world task or request, understands it, uses relevant tools and data, and executes a multi-step workflow. The agent should do more than generate text.

Possible Application Areas

- Civic services and community operations.
- Education, healthcare, accessibility, and productivity.
- Sustainability and other real-world challenges.
- Open-source project workflows.

Core Challenge: Use an open-source or open-weight AI model as a meaningful part of the agent. Demonstrate appropriate tool use, decision-making, workflow orchestration, and failure handling.

PS 04 — Build the AI System Behind the AI

Build an open-source AI infrastructure component, model harness, agent skill, or evaluation system that improves how developers build, test, or use open-source AI.

Possible Directions

- An AI agent skill or model evaluation framework.
- A model-routing system or tool-use/function-calling harness.
- An AI workflow reliability layer or privacy/local-inference framework.
- A system for comparing open-weight models or an open-source AI developer tool.

Core Challenge: Solve a genuine technical problem in the open-source AI ecosystem rather than simply wrapping an existing model in a user interface. A model-harness project must include an original implementation or meaningful changes to an existing open-source harness. Agent-skill submissions must follow the Agent Skills open standard.

OFFICIAL MLH CHALLENGE REQUIREMENTS

- **Best Use of Gemma 4:** Gemma 4 must be meaningfully used in the project. Consider its text-and-image capabilities or build a focused tool for learning, creativity, productivity, or community use.
- **Best Open-Source AI Project:** Open-source or open-weight AI must be an important part of how the project works.
- Publish the project in a **public GitHub repository** with an **open-source license**. Include a README explaining what the project does, how to run or try it, and the model and key dependencies used.
- If submitting an agent skill, follow the Agent Skills open standard. If submitting a model harness, include original implementation or meaningful improvements.

GENERAL PARTICIPANT GUIDELINES

- **Team size:** 2–4 members. Each participant must be registered individually on the official MLH event portal.
- **Build during the Hack Day:** Develop the final solution during the designated event period. Do not submit a pre-built project as the Hack Day solution.
- **Open-ended problem solving:** The problem statements are intentionally broad. Teams choose their use case, architecture, models, tools, APIs, databases, cloud services, and workflow frameworks.
- **Demonstrate real functionality:** Be prepared to show a working prototype and explain the role of AI, architecture, data sources, and key technical decisions.
- **Responsible use:** Protect credentials and API keys. Do not expose private or sensitive data. Clearly disclose limitations and avoid unsupported claims about your system.
- **Code of Conduct:** Follow the MLH Code of Conduct, respect fellow participants, mentors, judges, volunteers, and organizers, and comply with all MSRIT campus rules.
- **Judging:** Ensure your project is ready for the scheduled submission and demo. Be prepared to explain the problem, technical approach, implementation, feasibility, and impact.

TECHNOLOGY CHOICE

There is no prescribed tech stack. Teams may use frameworks, APIs, databases, workflow/orchestration tools, cloud services, and open-source libraries where appropriate. AWS usage in PS 01 is encouraged but not mandatory. Choose tools that support the solution; the technology itself should not be the solution.

OFFICIAL RESOURCES

MLH Event & Challenges: [View official challenge page](#)

MLH Open-Source AI Requirements: [View MLH Open-Source AI guidelines](#)

MLH Code of Conduct: [Read the Code of Conduct](#)

Optional AWS/Kiro resource: [Kiro Workshop — Kicking Things Off](#)

Please refer to the official MLH challenge page for the latest challenge-specific requirements. Where a challenge has additional requirements, those requirements take precedence.